KTU CST204 - DBMS Module 3 Solved Answers

June 2023

Q15(a) [4 Marks] - What is an assertion? How is it different from a trigger?

- Assertion: A condition that must always hold true in the database.

- Difference from Trigger:

  * Assertion: Global constraint, always enforced.

  * Trigger: Executes a procedure based on events like INSERT or UPDATE.

Q15(b) [10 Marks] - SQL Queries on EMPLOYEE and DEPARTMENT:

i. Employees in 'Research' department:

SELECT E.Fname, E.Lname, E.Address FROM EMPLOYEE E, DEPARTMENT D WHERE E.Dno = D.Dnumber AND D.Dname = 'Research';

ii. Employee and supervisor names:

SELECT E.Fname AS Employee, S.Fname AS Supervisor FROM EMPLOYEE E JOIN EMPLOYEE S ON E.SuperSSN = S.SSN;

iii. Employees working on all projects of department 5:

```
SELECT E.Fname, E.Lname FROM EMPLOYEE E WHERE NOT EXISTS (
  SELECT P.Pnumber FROM PROJECT P WHERE P.Dnum = 5
  EXCEPT
  SELECT W.Pno FROM WORKS_ON W WHERE W.ESSN = E.SSN );
```

iv. Projects involving or managed by 'Smith':

SELECT DISTINCT P.Pnumber FROM PROJECT P JOIN DEPARTMENT D ON P.Dnum = D.Dnumber JOIN EMPLOYEE E ON (P.Dnum = E.Dno AND (E.Lname = 'Smith' OR D.MgrSSN = E.SSN));

v. SSNs working on project 1, 2 or 3:

SELECT DISTINCT ESSN FROM WORKS_ON WHERE Pno IN (1, 2, 3);

Q16(a) [6 Marks] - Record size and indexing:

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- Record Size = 116 bytes.
- Blocking Factor = floor(512 / 116) = 4 records/block.
- First-level index = 7500 entries, index block size = 15 bytes -> 221 blocks.
- Second-level index = 7 blocks. Total levels = 2.

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June 2024

Q15(a) [4 Marks] - Types of Joins:

- Inner Join: Only matching records.

- Left Join: All from left + matching from right.

- Right Join: All from right + matching from left.

- Full Join: All records from both tables.

Q15(b) [10 Marks] - SQL on bank database:

i. Customers in Thiruvananthapuram:

SELECT DISTINCT C.name FROM CUSTOMER C JOIN DEPOSITOR D ON C.customer_id = D.customer_id JOIN ACCOUNT A ON D.account_number = A.account_number JOIN BRANCH B ON A.branch_name = B.branch_name WHERE B.branch_city = 'Thiruvananthapuram';

ii. Branches with assets > 100000:

SELECT branch_name FROM BRANCH WHERE assets > 100000;

iii. Customers with >1 account:

SELECT customer_id FROM DEPOSITOR GROUP BY customer_id HAVING COUNT(account_number) > 1;

iv. Branch name and total assets:

SELECT branch_name, SUM(assets) FROM BRANCH GROUP BY branch_name;

v. Name of customer deposited in account 1001:

SELECT C.name FROM CUSTOMER C JOIN DEPOSITOR D ON C.customer_id = D.customer_id WHERE D.account_number = 1001;

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December 2024

Q15(a) [4 Marks] - Multi-level Indexing:

- Used when index is too large for memory.

- Several levels of indexes until top-level index fits in one block.

Q15(b) [10 Marks] - SQL on university schema:

i. Students in DBMS:

SELECT S.Name FROM STUDENT S JOIN ENROLL E ON S.RollNo = E.RollNo JOIN COURSE C ON E.CourseId = C.CourseId WHERE C.Title = 'DBMS';

ii. Courses taught by CSE instructors:

SELECT C.Title FROM COURSE C JOIN INSTRUCTOR I ON C.Dept = I.Dept WHERE I.Dept = 'CSE';

iii. Students with grade A:

SELECT S.Name FROM STUDENT S JOIN ENROLL E ON S.RollNo = E.RollNo WHERE E.Grade = 'A';

iv. Instructors teaching >3 courses:

SELECT I.Name FROM INSTRUCTOR I JOIN COURSE C ON I.Dept = C.Dept GROUP BY I.InstructorId, I.Name HAVING COUNT(C.CourseId) > 3;

v. Students not enrolled in any course:

SELECT Name FROM STUDENT WHERE RollNo NOT IN (SELECT RollNo FROM ENROLL);

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January 2024

Q15(a) [4 Marks] - Trigger and Types:

- Trigger: Procedure that runs automatically when an event happens on a table.

- Types: BEFORE, AFTER, INSTEAD OF.

Example:

CREATE TRIGGER check_salary BEFORE INSERT ON EMPLOYEE FOR EACH ROW BEGIN IF NEW.salary < 0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Invalid salary'; END IF; END;

Q15(b) [10 Marks] - Similar SQL Queries (University schema). Refer to Dec 2024 section.

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June 2022

Q15(a) [4 Marks] - View vs Table:

- View: Virtual table, stores SQL query.

- Table: Physical storage of data.

Q15(b) [10 Marks] - SQL Queries (EMPLOYEE/PROJECT). Similar to June 2023.

Q16(a) [6 Marks] - Index calculation. Same as June 2023.

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July 2021

Q15(a) [4 Marks] - Normalization:

- 1NF: Atomic values only.

- 2NF: No partial dependency.

- 3NF: No transitive dependency.

- BCNF: Every determinant is a candidate key.

Q15(b) [10 Marks] - SQL Queries (SALES, PRODUCT). Example:

SELECT * FROM PRODUCTS WHERE Price > 1000;